

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2016-17

Paper: BCS-503 — Discrete Mathematical Structures

Time: Three Hours

Full Marks: 70

*Instructions: Answer any five questions. Terms/Symbols used have their standard meanings.***Question 1.** Define the following terminologies and explain with appropriate examples: $[2 \times 7 = 14]$

- (a) Poset
- (b) Regular Graph
- (c) Clique
- (d) Fundamental Circuit
- (e) Bipartite Graph
- (f) Minimal Covering
- (g) De Morgan's Law

Question 2.

- (a) Explain partition of a set and equivalence classes with a suitable example. [3]

- (b) What is a Hasse Diagram? Draw the Hasse Diagram for the relation R on set $A = \{a, b, c, d, e\}$ where R is given as:

$$R = \{\langle a, a \rangle, \langle a, c \rangle, \langle a, d \rangle, \langle a, e \rangle, \langle b, b \rangle, \langle b, c \rangle, \langle b, d \rangle, \langle b, e \rangle, \langle c, c \rangle, \langle c, d \rangle, \langle c, e \rangle, \langle d, d \rangle, \langle e, e \rangle\}$$

[4]

- (c) Find the inverse of the function $f(x) = 4e^{x+2}$. [3]

- (d) A mobile manufacturing company assembled 30 mobiles. The features available are Bluetooth, Wi-Fi and 4G network. It is known that 15 of the mobiles have Bluetooth, 8 of them have Wi-Fi, 6 of them have 4G network, and 3 of them have all three features. Find at least how many mobiles do not have any features at all. [4]

Question 3.

- (a) For an algebraic system $(A, \vee, \wedge)$ defined by a lattice $(A, \leq)$, show that: [5]

(i) $a \leq a \vee b, a \wedge b \leq a$

(ii) $a \vee (a \wedge b) = a, a \wedge (a \vee b) = a$ where $a, b \in A$.

- (b) Define Boolean lattice and Boolean algebra. Prove that an element in a Boolean lattice has a unique complement. [5]

- (c) Given a function $f : \{0, 1\}^3 \rightarrow \{0, 1\}$ defined as:

$$\begin{array}{ll} f(0, 0, 0) = 1, & f(0, 0, 1) = 0, \\ f(0, 1, 0) = 1, & f(0, 1, 1) = 0, \\ f(1, 0, 0) = 0, & f(1, 0, 1) = 0, \\ f(1, 1, 0) = 0, & f(1, 1, 1) = 1. \end{array}$$

Express the function in disjunctive normal form and conjunctive normal form of Boolean expression. [4]

Question 4.

- (a) Determine whether each of the following compound propositions is a tautology or not: [3]

(i) $(\neg p \wedge \neg q) \rightarrow (p \rightarrow q)$

(ii) $(p \rightarrow q) \rightarrow (p \wedge \neg q)$

(iii) $[p \wedge (p \rightarrow \neg q)] \rightarrow q$

- (b) Show that $\neg(p \vee (\neg p \wedge q)) \equiv \neg p \wedge \neg q$ and $\neg \forall x(P(x) \rightarrow Q(x)) \equiv \exists x(P(x) \wedge \neg Q(x))$ without truth table. [4]

- (c) Solve the following recurrence relations: [7]

(i) $4a_r - 20a_{r-1} + 17a_{r-2} - 4a_{r-3} = 0$

(ii) $a_r + 5a_{r-1} + 6a_{r-2} = 3^r - 2r + 1$

Question 5.

- (a) Prove that the number of vertices of odd degree in a graph is always even. [3]

- (b) Draw the graphs of the following types: [5]

(i) Graph is both Hamiltonian and Eulerian.

(ii) Graph is Eulerian but not Hamiltonian.

(iii) Graph is Hamiltonian but not Eulerian.

- (c) What is meant by center of a graph? Prove that a tree may have one or two centers. [2]

- (d) What will be the minimum and maximum possible height of a binary tree with n vertices? [2]

- (e) What are the Kuratowski's two graphs? Also draw these two graphs. [2]

Question 6.

- (a) How to prove that a graph is non-planar? Show with an example. [4]

- (b) What is geometric dual of a graph? If r and μ denote the rank and nullity of a graph G , and if r^* and μ^* denote the same for the dual graph G^* , then prove that $r^* = \mu$ and $\mu^* = r$. [4]

- (c) Let $G = (V, E)$ be a connected weighted and undirected graph, where $V = \{a, b, c, d, e, f, g\}$ is the set of vertices and

$$E = \{\langle(a, b), 3\rangle, \langle(a, c), 2\rangle, \langle(b, d), 1\rangle, \langle(c, d), 5\rangle, \langle(c, f), 7\rangle, \langle(d, e), 3\rangle, \langle(d, f), 6\rangle, \langle(e, g), 8\rangle, \langle(f, g), 4\rangle\}$$

is the set of edges along with weights. Find the minimal spanning tree of G using Kruskal's and Prim's algorithms with demonstration of steps in both methods. [6]

Question 7.

- (a) What is chromatic partitioning? Explain with an example. [3]
- (b) What is chromatic polynomial? Find the chromatic polynomial of a complete graph of n vertices and a tree of n vertices. [5]
- (c) Explain the reduced incidence matrix A_f , fundamental circuit matrix B_f , and fundamental cutset matrix C_f of a connected graph. Use a common graph and common spanning tree to explain these matrices. What is the relationship among them? [6]